Try it at the table

How a Sidewalk Is Made and How It Works

Gather your things, then try each one. Guess first. Then look closely and write or draw what you saw.

What you need

- ☐ Paper and pencil
- ☐ Paper cup
- ☐ Flour
- ☐ Water
- ☐ Dry rice
- ☐ Stick of sidewalk chalk
- ☐ Two books
- ☐ Graham cracker

TRY 1 What is a sidewalk made of?

1. Look closely at a sidewalk square outside, or at a photo of one.
2. Find tiny stones, speckles and a groove.
3. Draw one square on paper and mark where the groove runs.

I THINK…

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I SAW…

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TRY 2 How does wet concrete turn hard?

1. Stir two spoonfuls of flour with a little water in a paper cup to make a thick paste.
2. Stir in a handful of dry rice so the paste coats every grain.
3. Leave the cup for a day, then tap it and feel how hard it got.
4. Flour paste only dries out, but real concrete hardens by a chemical reaction.

I THINK…

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I SAW…

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TRY 3 Why is a sidewalk strong, and what holds it up?

1. Stand a stick of sidewalk chalk on end and press down on top with one finger to see that it holds.
2. Now lay the chalk across two books and press gently in the middle until it snaps.
3. Chalk is a stand-in for concrete: strong when squeezed, weak when bent.

I THINK…

I SAW…

TRY 4 Why does a sidewalk have grooves?

1. Look at the lines on a graham cracker.
2. Break one cracker along a line.
3. Then break another across the middle where there is no line.
4. See which break is neat and straight.

I THINK…

I SAW…